



MERN Stack Internship – Day 1

Introduction to Web Development

What is Web Development?



Building Websites & Web Apps

Creating interactive digital experiences that run in browsers, from simple landing pages to complex applications.



Frontend vs Backend

Frontend is what users see and interact with. Backend handles data processing and storage behind the scenes.



Full Stack Developer

Master of both frontend and backend technologies, capable of building complete web applications from start to finish.

Meet the MERN Stack

A powerful combination of four technologies that work together to create modern web applications.



MongoDB

A flexible NoSQL database that stores data in JSON-like documents, making it easy to work with.



Express.js

A lightweight framework for building backend APIs and handling server-side logic efficiently.



React.js

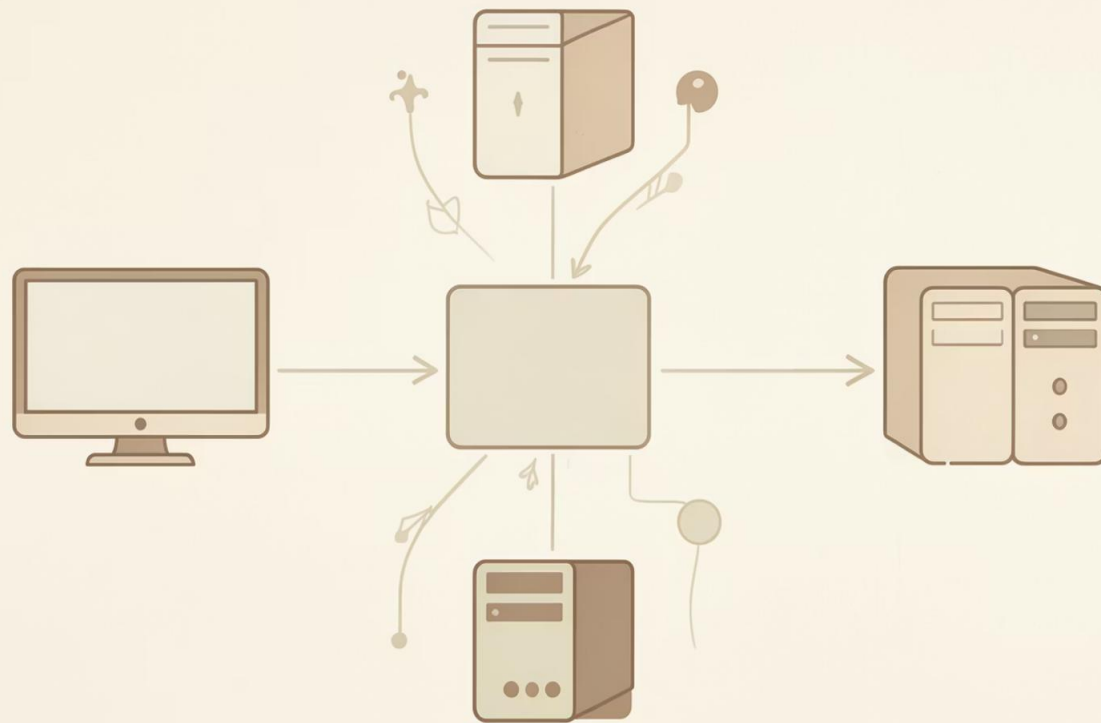
A popular frontend library for building dynamic user interfaces with reusable components.



Node.js

A JavaScript runtime that lets you run code outside the browser, powering the server.

How the Internet Works



The Basic Flow

When you type a URL in your browser, a request travels to a server. The server processes your request, retrieves the needed information, and sends back a response with the webpage content.

- **Client:** Your browser makes requests
- **Server:** Computer that processes requests
- **HTTP:** The language they use to communicate
- **Response:** Webpage content sent back

Setting Up Your Development Environment

Before we start coding, we need to install the essential tools that will power your development journey.

01

Visual Studio Code

A powerful, free code editor with features like syntax highlighting, debugging tools, and thousands of extensions to supercharge your coding.

02

Node.js

Installs the runtime environment and package manager (npm) needed to run JavaScript on your computer and install dependencies.

03

Chrome Browser

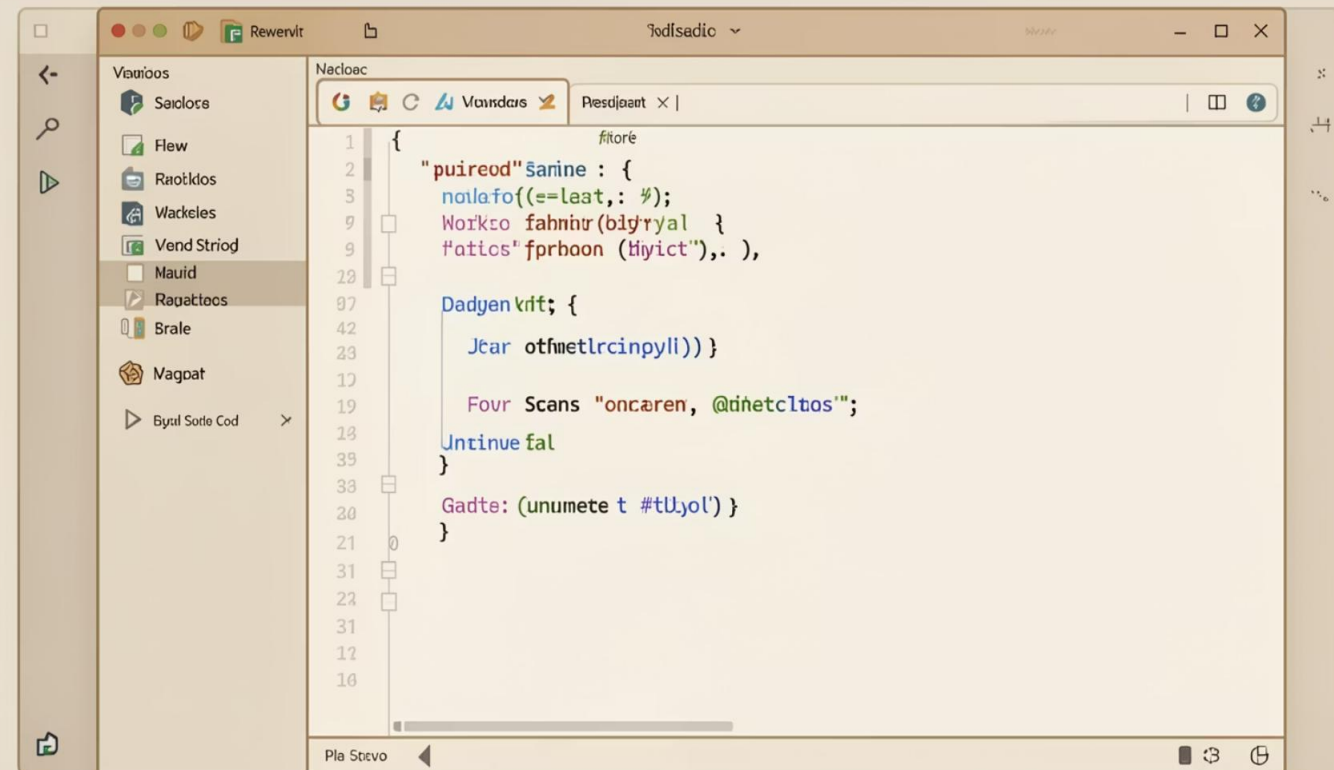
Your testing ground with developer tools for inspecting elements, debugging code, and analyzing network activity.

Installing Visual Studio Code

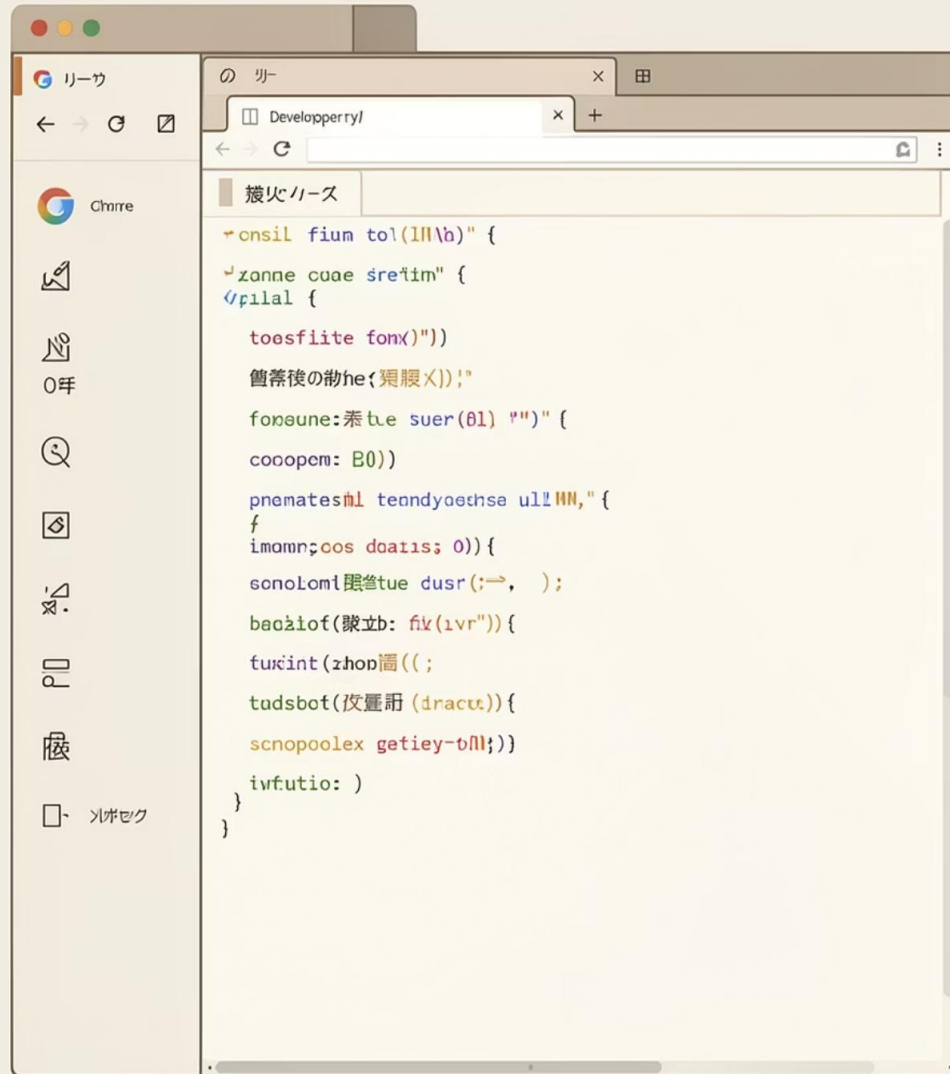
Step-by-Step Installation

1. Visit code.visualstudio.com
2. Download the installer for your operating system
3. Run the installer and follow the prompts
4. Launch VS Code and explore the interface
5. Install recommended extensions for web development

Pro Tip: Install the Prettier and ESLint extensions for automatic code formatting and error detection!



Installing Node.js & Chrome



Node.js Installation

- Download from nodejs.org
- Choose the LTS (Long Term Support) version
- Run installer and accept default settings
- Verify installation in terminal: `node -v`

Chrome Browser

- Download from chrome.google.com
- Install and set as default browser
- Open DevTools with F12 or Ctrl+Shift+I
- Explore Elements, Console, and Network tabs

Creating Your First HTML File



Let's Code!

Create a new file called `index.html` and add this basic structure:

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Page</title>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>Welcome to web
development!</p>
</body>
</html>
```

Save the file, then open it in Chrome by right-clicking and selecting "Open with Chrome". You've just created your first webpage!

Homework for Day 2

1 Install All Development Tools

Complete installation of VS Code, Node.js, and Chrome on your computer. Verify each installation works by opening them and checking version numbers.

2 Create Your First HTML File

Build a simple HTML page with a heading, paragraph, and list. Experiment with adding images and links. Save it as `index.html` and open it in Chrome.

3 Explore Developer Tools

Open Chrome DevTools and inspect elements on popular websites. Right-click any element on a webpage and select "Inspect" to see the HTML structure.

You're Ready to Start!

Key Takeaways

- Web development combines frontend and backend
- MERN stack uses MongoDB, Express, React, Node.js
- Client sends requests, server sends responses

What's Next

- Day 2: Deep dive into HTML & CSS basics
- Learn to style your first webpage
- Start building interactive elements

Welcome to your web development journey! You now have the foundation and tools to start creating. See you tomorrow! 🚀

"WEB DEVELOPMENT JOURNEY,
CELEBRATING AND ACHIEVING
YOUR GOALS AND MISSIONS"

